



Prototype Development Plan

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Executive Summary

This document outlines the full development plan for a prototype Explainable Artificial Intelligence (XAI) system for credit risk scoring. The system combines supervised ensemble learning — specifically Random Forest and Gradient Boosting — with SHAP (Shapley Additive Explanations) to provide both highly accurate loan default predictions and human-readable decision transparency. A 3D React frontend renders risk scores and SHAP outputs as interactive three-dimensional visualisations, directly addressing the 'black box' problem identified in the research proposal. The plan follows the CRISP-DM methodology across six structured phases over fourteen weeks.

1. Technology Stack

Layer	Technology	Purpose	Key Libraries
Frontend — 3D	React + React Three Fiber	3D interactive UI	Three.js, Drei, Framer Motion, Leva
UI & Styling	Tailwind CSS + shadcn/ui	Component design system	shadcn/ui, Recharts (2D fallback)
Backend API	FastAPI (Python 3.11+)	REST prediction endpoints	Pydantic, Uvicorn, pytest
ML & XAI	Scikit-learn + SHAP	Model training & explanation	Random Forest, GBM, TreeExplainer
Data Layer	Pandas + SQLite	Data pipeline & persistence	imbalanced-learn, Joblib
DevOps	Docker + Render	Containerised deployment	Docker Compose, GitHub Actions, Vite

React Three Fiber (R3F) is chosen over Streamlit because it integrates declaratively with React state — when the FastAPI backend returns a new SHAP payload, the 3D scene updates reactively without manual Three.js scene graph management. FastAPI is preferred over Flask for its async support, automatic OpenAPI documentation, and Pydantic-based validation.

2. 3D Visualisation Components

Three signature 3D components differentiate this prototype from a standard Streamlit dashboard and directly address the transparency requirements identified in the research proposal:

Component	Description	XAI Purpose
Risk Orb	A pulsing Three.js sphere on the results screen. Its scale, color, and rotation are dynamically updated based on the model's predicted risk score and confidence level.	Visualizes the model's overall risk assessment and confidence, providing a high-level summary of the prediction's reliability.
3D SHAP Bar Forest	Global feature importance rendered as extruded 3D bars in a forest-like arrangement. Each bar's height corresponds to the SHAP value for a specific feature across the dataset.	Provides an intuitive, spatial representation of feature importance, allowing users to quickly identify the most influential factors in the model's decision.
Animated Waterfall	Local SHAP values for a single applicant assembled as floating, interconnected blocks. Each block represents the contribution of a specific feature to the final prediction for that individual.	Enables users to drill down into the model's reasoning for a specific case, showing how individual features contribute to the final risk score.
Counterfactual Panel	Slider panel that re-queries the API with modified inputs in real-time. It visualizes how changing specific features (e.g., income, credit score) would alter the model's prediction.	Implements the 'What-If' analysis, helping users understand the sensitivity of the model's output to changes in input features and identifying potential paths to a more favorable outcome.

3. Build Phases — CRISP-DM Aligned

The project follows the six-phase CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology as specified in Section 1.8 of the proposal, with an additional DevOps deployment phase.

Phase 1: Business & Data Understanding · Weeks 1–2

- Define loan officer user stories and map to system requirements.
- Download and audit the German Credit Dataset (1,000 records, 20 features) and/or LendingClub open data.
- Conduct exploratory data analysis — class imbalance check, missing value profiling, correlation heatmaps.
- Document feature semantics: income, loan duration, employment history, age, credit history.
- Confirm regulatory and fairness-relevant demographic columns for the bias analysis (Specific Objective 4).

Deliverable: *EDA Jupyter notebook + feature dictionary document*

Phase 2: Data Preparation Pipeline · Weeks 3–4

- Impute missing values; encode categoricals via Scikit-learn Pipeline (OrdinalEncoder / OneHotEncoder).
- Apply SMOTE via imbalanced-learn to address class imbalance in default labels.
- Feature scaling (StandardScaler) and stratified 80/20 train-test split.
- Build a reusable preprocess.py module called identically at training and inference time.
- Persist the fitted pipeline with Joblib to ensure reproducibility across environments.

Deliverable: *preprocess.py module + saved pipeline artifact (.joblib)*

Phase 3: ML Model Training & SHAP Integration · Weeks 5–7

- Train Random Forest and Gradient Boosting classifiers; tune hyperparameters with RandomizedSearchCV.
- Evaluate with AUC-ROC, F1-score, precision/recall — compare both models to address Research Question 2.
- Integrate shap.TreeExplainer — compute global SHAP summary plots for feature importance ranking.
- Compute local SHAP values per prediction — waterfall plots for individual loan decisions.
- Fairness audit: compare false positive rates across demographic subgroups (age, gender if present) per Cui et al. (2022).
- Serialise champion model + SHAP explainer with Joblib for API consumption.

Deliverable: *Trained model + SHAP explainer + fairness report notebook*

Phase 4: FastAPI Backend · Weeks 7–9

- Scaffold FastAPI app with /predict endpoint — accepts JSON applicant payload, returns risk score + SHAP values.
- Add /health, /model-info, and /global-importance endpoints.
- Pydantic models for strict input validation (loan amount ranges, categorical enumerations).
- Load persisted pipeline and model at startup via lifespan context — single load, reused per request.
- Enable CORS for local React dev server; add basic rate limiting.
- Write pytest unit tests for each endpoint and edge cases (missing fields, out-of-range values).

Deliverable: *Tested FastAPI app with auto-generated OpenAPI docs at /docs*

Phase 3: 3D React Frontend · Weeks 9–12

- Bootstrap Vite + React + Tailwind project; install React Three Fiber, @react-three/drei, Framer Motion.
- Build Risk Orb — pulsing Three.js sphere whose colour and particle density encode the risk score.
- Build 3D SHAP Bar Forest — extruded bar chart rendered in R3F showing global feature importance.
- Build Animated Waterfall — local SHAP slabs assembled in sequence via Framer Motion stagger.
- Build loan officer input form (shadcn/ui) with live validation; POST to /predict, parse response, update 3D scene.
- Build Counterfactual Panel — real-time slider re-query implementing Miller (2019) counterfactual theory.
- Build Fairness Dashboard page — 2D grouped bar charts (Recharts) comparing approval rates across groups.

Deliverable: *Functional React app with 3D visualisations connected to live FastAPI backend*

Phase 4: Evaluation, Fairness & Deployment · Weeks 12–14

- End-to-end user testing with sample applicant profiles; validate SHAP outputs match model behaviour.
- Compile final model evaluation table: Accuracy, F1-Score, AUC-ROC for both RF and GBM.
- Write interpretability evaluation — qualitative assessment of SHAP explanations for non-technical users.
- Containerise full stack with Docker Compose (frontend + backend as separate services).
- Deploy to Render (free tier) — backend as Web Service, frontend as Static Site.
- Record demo video walkthrough for submission; finalise written report.

Deliverable: *Live deployed prototype URL + final research report*

4. 14-Week Project Timeline

Phase	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Business & Data Understanding														
Data Preparation Pipeline														
ML Training & SHAP Integration														
FastAPI Backend														
3D React Frontend														
Evaluation & Deployment														

Phases 3 and 4 overlap in weeks 7–9: the FastAPI backend development begins while ML model training is being finalised. The frontend (Phase 5) can start with mocked API responses in week 9 before the backend is complete.

5. Evaluation Framework

5.1 Model Performance Metrics

Metric	Definition	Target	Addresses
AUC-ROC	Area under the Receiver Operating Characteristic curve — measures overall discriminative ability.	> 0.80	Specific Obj. 1
F1-Score	Harmonic mean of precision and recall — balances detecting defaulters vs. false positives.	> 0.75	Specific Obj. 4
Accuracy	Proportion of correct classifications across all applicants.	> 0.80	Research Q. 4
Precision	Of predicted defaults, what fraction actually defaulted.	Report only	Research Q. 1
Recall	Of actual defaults, what fraction the model correctly identified.	Report only	Research Q. 2

5.2 Fairness Analysis

A fairness audit is conducted per Cui et al. (2022) to detect potential algorithmic bias inherited from historical lending data. The analysis compares false positive rates (incorrectly flagging a creditworthy applicant as a defaulter) across demographic subgroups available in the dataset. SHAP dependence plots are used to identify whether any demographic feature disproportionately drives predictions. Results are surfaced on the Fairness Dashboard page of the frontend.

5.3 Interpretability Evaluation

Addressing Research Question 3, the interpretability of SHAP outputs is assessed qualitatively: sample applicant profiles are evaluated and their SHAP waterfall outputs reviewed for whether the explanation is understandable to a non-technical user (a loan officer with no ML background). The counterfactual panel is tested to confirm that modifying the top SHAP features produces score changes consistent with the direction of the SHAP value.

6. Alignment with Research Proposal

Proposal Element	Implementation
Specific Obj. 1: Preprocess & identify key features	Phase 2 — Scikit-learn Pipeline with EDA notebook identifying top default predictors.
Specific Obj. 2: Train RF & GBM, validate	Phase 3 — RandomizedSearchCV tuning; AUC-ROC & F1 comparison table across both models.
Specific Obj. 3: Integrate SHAP for global & local explanations	Phase 4 — SHAP explainer summary plots (global); Phase 5 — 3D waterfall (local).
Specific Obj. 4: Evaluate performance & interpretability	Phase 6 — Full metrics report + qualitative interpretability assessment + fairness audit.
Research Q. 1: Key default-risk features	EDA heatmaps + SHAP global importance bar forest identify and rank top predictors.
Research Q. 2: RF vs. GBM accuracy	Side-by-side evaluation table with AUC-ROC, F1, precision, recall for both models.
Research Q. 3: SHAP explaining individual decisions	Animated waterfall + counterfactual panel tested with non-technical user scenarios.
Research Q. 4: XAI accuracy without transparency loss	High-performing model AUC-ROC > 0.80 combined with auditable SHAP outputs per prediction.
Gap 1: Functional software prototype	Full-stack prototype (FastAPI + React R3F) deployed to a live public URL on Render.
Gap 2: Local/student context	Prototype framed around Kenyan mobile lending context; datasets are open-source.
Gap 3: Holistic accuracy + explainability	Both quantitative metrics and qualitative interpretability evaluations are included.

7. Key References

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